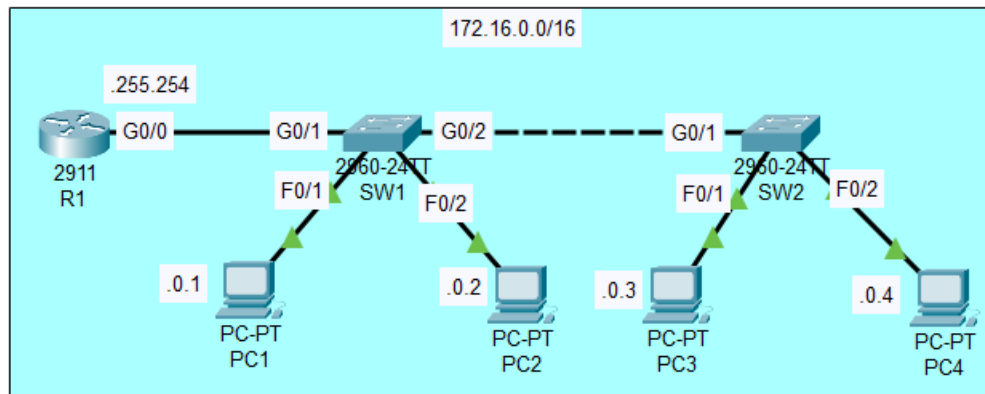




Topology

1. Configure the hostname of R1, SW1, and SW2.

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#hostname R1

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW1

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname SW2
```

2. Configure the appropriate IP addresses on R1, PC1, PC2, PC3, PC4

```
R1(config)#int g0/0
R1(config-if)#ip address 172.16.255.254 255.255.0.0
```

PC1

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☐ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0001.6484.DDAA

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.16.0.1

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local Address: FE80::201:64FF:FE84:DDAA

PC2

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0030.F21C.46E3

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.16.0.2

Subnet Mask 255.255.0.0

IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local Address: FE80::230:F2FF:FE1C:46E3

PC3

Physical **Config** Desktop Programming Attributes

GLOBAL

Settings

Algorithm Settings

INTERFACE

FastEthernet0

Bluetooth

FastEthernet0

Port Status ☒ On

Bandwidth ☒ 100 Mbps ☐ 10 Mbps ☒ Auto

Duplex ☐ Half Duplex ☒ Full Duplex ☒ Auto

MAC Address 0003.E442.5E07

IP Configuration

☐ DHCP

☒ Static

IPv4 Address 172.16.0.3

Subnet Mask 255.255.0.0

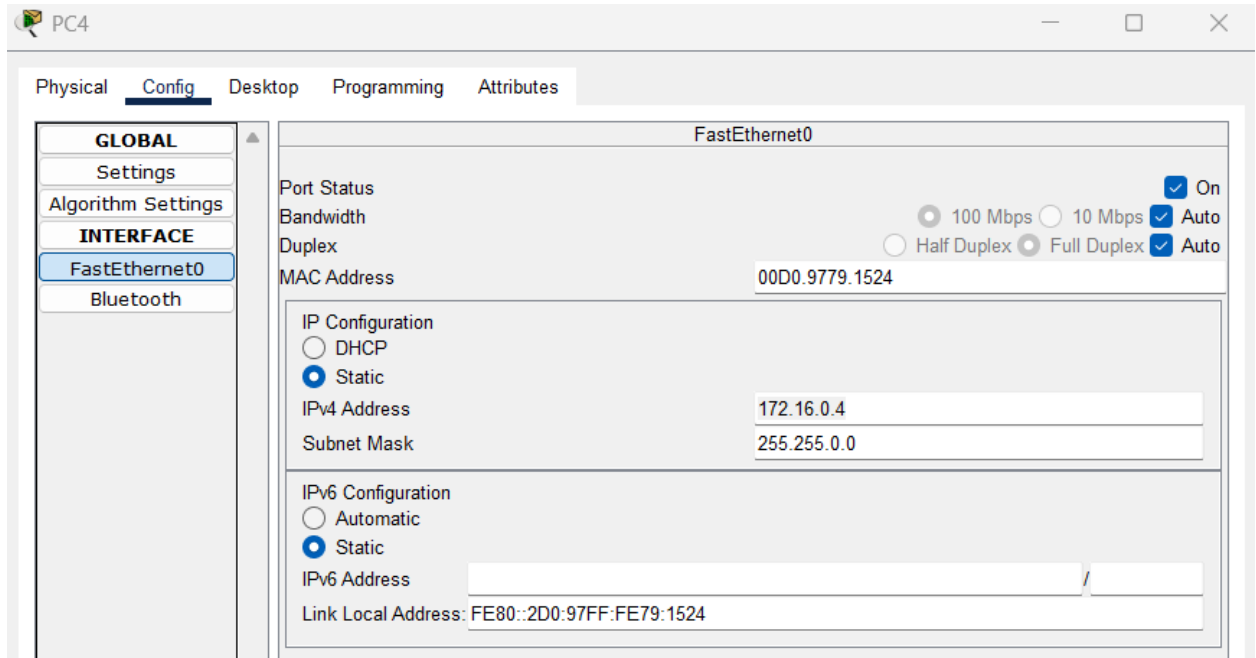
IPv6 Configuration

☐ Automatic

☒ Static

IPv6 Address /

Link Local Address: FE80::203:E4FF:FE42:5E07



3. Manually configure the speed and duplex on interfaces connected to other networking devices (not end hosts)

```
R1(config-if)#speed 1000
R1(config-if)#duplex full
SW1(config)#int g0/0
%Invalid interface type and number
SW1(config)#int g0/1
SW1(config-if)#speed 1000
SW1(config-if)#duplex full
SW1(config-if)#int g0/2
SW1(config-if)#speed 1000
SW1(config-if)#duplex full
SW1(config-if)#
%LINK-3-UPDOWN: Interface GigabitEthernet0/2, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down

SW2(config)#int g0/1
SW2(config-if)#speed 1000
SW2(config-if)#duplex full
SW2(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```

4. Configure appropriate descriptions on each interface

```
R1(config)#int g0/0
R1(config-if)#description ## to SW1 ##
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
SW1(config)#int g0/1
SW1(config-if)#description ## to R1 ##
SW1(config-if)#int g0/2
SW1(config-if)#description ## to SW2 ##
SW1(config)#int range f0/1 - 2
SW1(config-if-range)#description ## to end hosts ##
```

```

SW2(config-if)#description ## to SW1 ##
SW2(config-if)#
SW2(config)#int range f0/1 - 2
SW2(config-if-range)#description ## to end hosts ##
SW2(config-if-range)#int range g0/2, f0/3 - 24
SW2(config-if-range)#description ## not in use ##
SW2(config-if-range)#

```

5. 5. Disable interfaces which are not connected to other devices.

```

SW1(config)#int range f0/3 - 24
SW1(config-if-range)#description ## not in use ##
SW1(config-if-range)#shutdown

```

```

%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down

```

```
SW2(config)#int range g0/2, f0/3 - 24
SW2(config-if-range)#shutdown
```

```
%LINK-5-CHANGED: Interface GigabitEthernet0/2, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/4, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/5, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/6, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/7, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/8, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/9, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/10, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/11, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/12, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/13, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/14, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/15, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/16, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/17, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/18, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/19, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/20, changed state to administratively down
%LINK-5-CHANGED: Interface FastEthernet0/21, changed state to administratively down
```

6. Show interface status

```

SW1(config)#do sh int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     ## to end hosts ## connected  1          auto    auto   10/100BaseTX
Fa0/2     ## to end hosts ## connected  1          auto    auto   10/100BaseTX
Fa0/3     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/4     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/5     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/6     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/7     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/8     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/9     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/10    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/11    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/12    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/13    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/14    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/15    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/16    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/17    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/18    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/19    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/20    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/21    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/22    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/23    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/24    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Gig0/1    ## to R1 ##    connected  1          a-full  a-100010/100BaseTX
Gig0/2    ## to SW2 ##   connected  1          a-full  a-100010/100BaseTX

SW2(config)#do show int status
Port      Name      Status      Vlan      Duplex  Speed  Type
Fa0/1     ## to end hosts ## connected  1          auto    auto   10/100BaseTX
Fa0/2     ## to end hosts ## connected  1          auto    auto   10/100BaseTX
Fa0/3     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/4     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/5     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/6     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/7     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/8     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/9     ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/10    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/11    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/12    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/13    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/14    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/15    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/16    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/17    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/18    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/19    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/20    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/21    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/22    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/23    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Fa0/24    ## not in use ## disabled 1          auto    auto   10/100BaseTX
Gig0/1    ## to SW1 ##   connected  1          a-full  a-100010/100BaseTX
Gig0/2    ## not in use ## disabled 1          auto    auto   10/100BaseTX

```

7. Save configuration changes

```

R1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

SW1#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]

```

```
SW2#copy running-config startup-config
Destination filename [startup-config]?
Building configuration...
[OK]
```